Vijay Kant Singh

Junior Software Developer | MERN Stack Developer

Dayal Bagh, Agra
vijaychaudhary1400@gmail.com
+91-7818994402

LinkedIn: www.linkedin.com/in/vijay-kant-singh | GitHub: https://github.com/VijayChaudhary2557

EDUCATION

Master of Computer Applications (MCA) 2024-2026
GLA University, Mathura

Bachelor of Science (Mathematics) 2021-2024
R.B.S. College, Agra

SUMMARY

Results-driven **Junior Software Developer** with hands-on experience in designing and developing enterprise-grade web and desktop applications using the **MERN Stack**. Experienced in building geospatial visualization platforms, ERP systems, UAV inventory management solutions, and Electron-based desktop software. Skilled in integrating REST APIs, working with GeoTIFF satellite imagery, interactive mapping, and GIS technologies. Passionate about developing scalable, high-performance applications and solving real-world business challenges through modern software engineering.

EXPERIENCE

1. **Agrocast Analytics Pvt. Ltd.** (Junior Software Developer) **- Feb 2026 - Present**
   - Developing enterprise-level web and desktop applications using React.js, Node.js, Express.js, MongoDB, and Electron.js.
   - Designing scalable REST APIs and responsive user interfaces.
   - Building GIS and geospatial applications using Leaflet, GeoTIFF datasets, and ISRO APIs.
   - Collaborating with cross-functional teams to deliver production-ready software.
   - Optimizing application performance and improving code quality.
   - Developed MERN Stack applications for internal business operations.
   - Integrated frontend applications with secure backend APIs.
   - Worked on GIS-based visualization using Leaflet, Google Earth Engine, and QGIS.

TECHNICAL SKILSS

- **Programming Language:** Javascript(ES6+), Java, Python
- **Frontend:** React.js, HTML, CSS, Bootstrap, Tailwind CSS,
- **Backend:** Node.js, Express.js REST API Development
- **Database:** MongoDB, MySQL
- **GIS & Geospatial Technologies:** Leaflet.js, Google Earth Engine, QGIS, GeoTIFF Processing, ISRO API Integration
- **Desktop Development:** Electron.js
- **Tools:** Git & GitHub, Postman, VS Code

PROJECTS

1. **Hydrview (React.js | Node.js | Express.js | MongoDB | Leaflet | ISRO APIs)**
   - Developed a geospatial web application to visualize GeoTIFF satellite imagery and soil moisture data.
   - Integrated GeoTIFF layers with **Leaflet** to render interactive satellite maps.
   - Configured a remote development environment using **VS Code SSH** to securely access ISRO APIs hosted on the company's Ubuntu server.

- Processed large GeoTIFF datasets using **QGIS** by cropping and resampling images for optimized web visualization.
- Improved visualization performance by optimizing high-resolution satellite imagery for browser-based rendering.

2. ERP Management System (React.js | Node.js | Express.js | MongoDB)

- Developed an ERP platform for managing projects, employees, and business operations.
- Built project assignment, employee work tracking, and quotation management modules.
- Implemented daily employee logs, attendance tracking, and payroll features including salary slips.
- Designed secure role-based access and REST APIs to streamline organizational workflows.

3. ThermaXK Software (Electron.js | React.js | Node.js | Express.js | MongoDB | Leaflet)

- Developed a cross-platform desktop application using **Electron.js** and the **MERN Stack**, packaged as a standalone **.exe** application.
- Built a thermal image processing workflow that generates **GeoTIFF** files by combining drone-captured **JPEG** and **BIN** file pairs.
- Integrated **Leaflet** to visualize generated thermal GeoTIFF maps, enabling intuitive temperature analysis.
- Implemented subscription-based features including **payment gateway integration**, **promo codes**, **free trial codes**, and license management.
- Designed secure REST APIs and responsive UI for efficient thermal data processing and user management.